

Setting the Standard for Automation™

Alarm Management

Current State and Direction for Alarm Management Guidelines

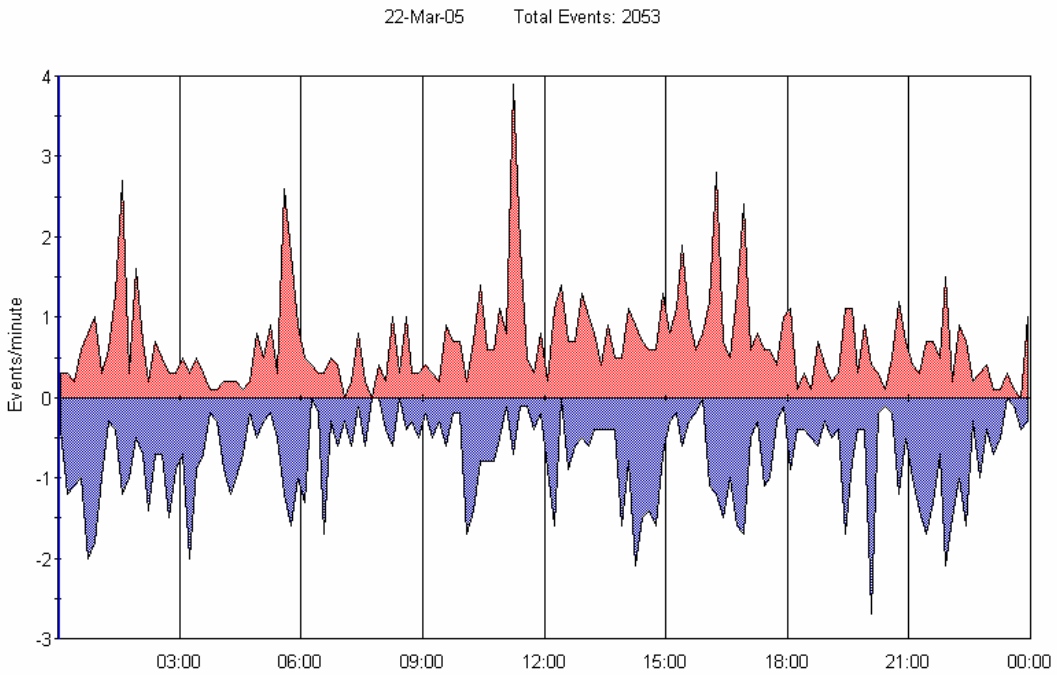
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Outline

- Background
- Common Problems of Alarm Management
- Alarm Management Lifecycle
- SP 18 Status
- Getting Started

Alarms Problems Today



- In most plants there are far more alarms to the operator than needed.
- Many alarm management features are not used.
- Alarm systems are not always monitored for performance.

Cost of Poor Alarm Management

- Estimated cost to US industry at over 20 billion dollars/year
- Recognized common problem in industry
- Often cited as contributing factors in industrial incidents
  - Milford Haven
  - Three Mile Island
  - Chernobyl
  - BP explosion
  - And so many more

Control Panel to Control Systems

- Distributed Control Systems have replaced panel control rooms.
- The number of tags, or data points has increased 100X.





- The space to display process information has decreased.
- The area of responsibility for operators has increased.

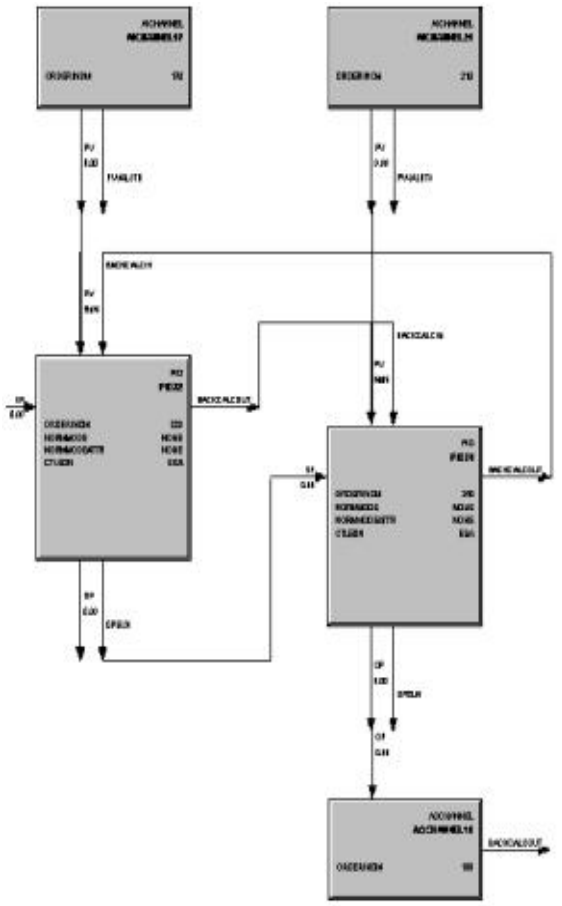
Decreasing display area per operator

Increasing point count per operator

Increasing Alarm Count

- Panel alarms were limited and expensive to add alarms.
- DCS alarms are built into to the tags, with up to 14 alarm limits.
- Many alarms are set because the are free.





Decreasing cost per alarm

Increasing alarms per point

Common Alarm Problems

1. Nuisance alarms
2. Stale alarms
3. Alarm Floods
4. Alarms without response
5. Alarms with the wrong priority
6. Out-of-Service alarms
7. Redundant alarms

Common Alarm Problems

1. Nuisance alarms
   - Alarms that trigger when no abnormal condition exists or when no operator action is required.
   - Desensitize the operator.
   - Reduce the response to real alarms.
   - Often caused by maintenance issues or improper limits.
2. Stale alarms
   - Alarms that remain in alarm for extended periods.
   - No operator action is required.
   - Do not clear after operator action has been taken.
   - Clutter the alarm system.

Common Alarm Problems

3. Alarm Floods
   - Multiple alarms in a short time, usually triggered by a single event, (>10 alarms in 10 minutes).
   - Dangerous problem with alarm systems.
   - Most complex alarm problem to solve.
   - Potential to overwhelm the operator.
4. Alarms without response
   - Cause and/or response not documented for the operator.
5. Alarms with the wrong priority
   - Priority not used consistently.
   - Invalidates the priority system.

Common Alarm Problems

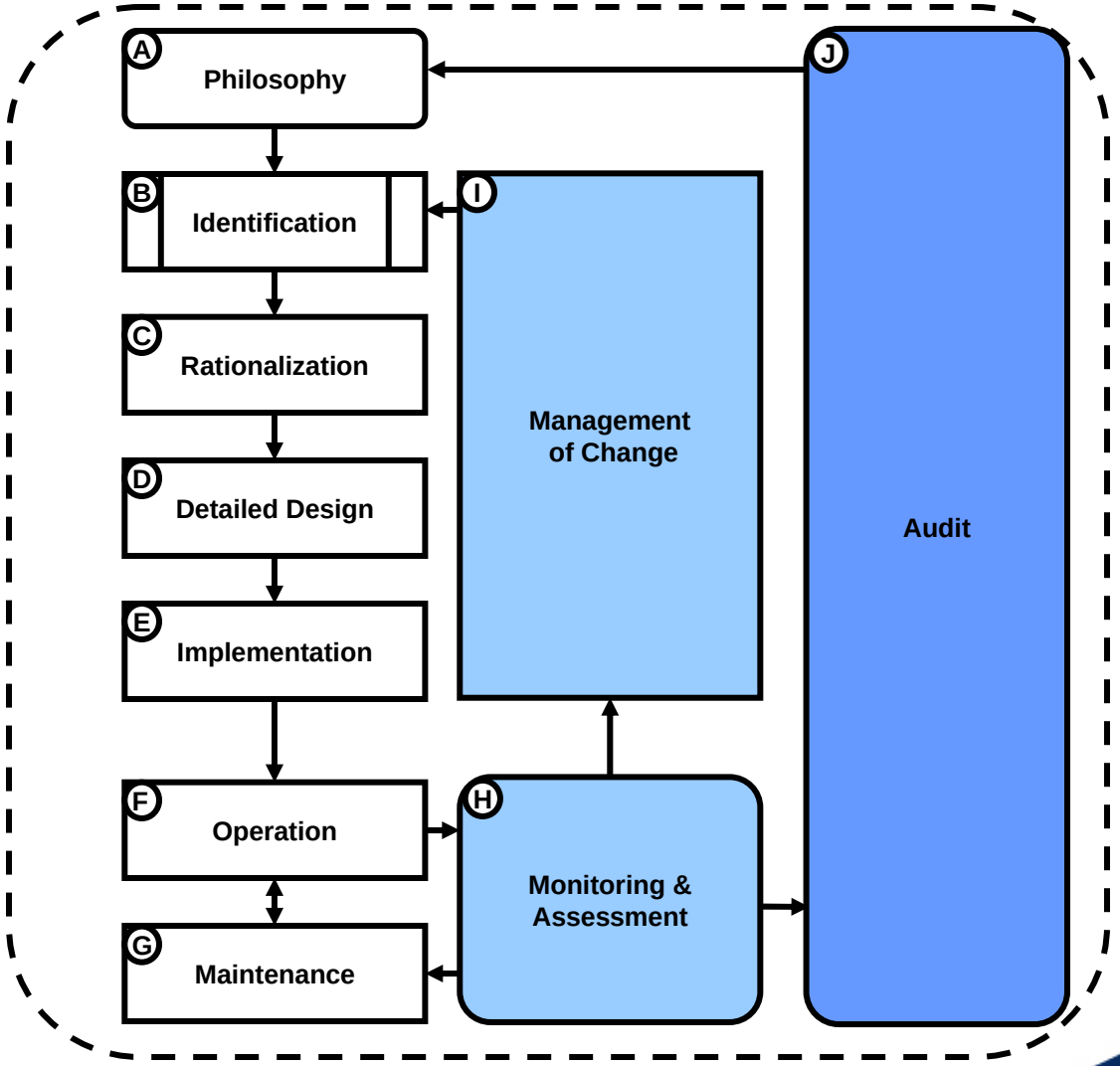
6. Out-of-Service alarms
   - Some alarms taken out of service with authorization.
   - Many alarms taken out of service without authorization.
7. Redundant alarms
   - Multiple alarms to indicate the same action.

Alarm Management Lifecycle

- ISA S18 draft lifecycle.
  - Includes practices for new facilities and existing plants.
  - Builds on the work of ASM and EEMUA.
  - Includes practices to solve the common alarm problems.

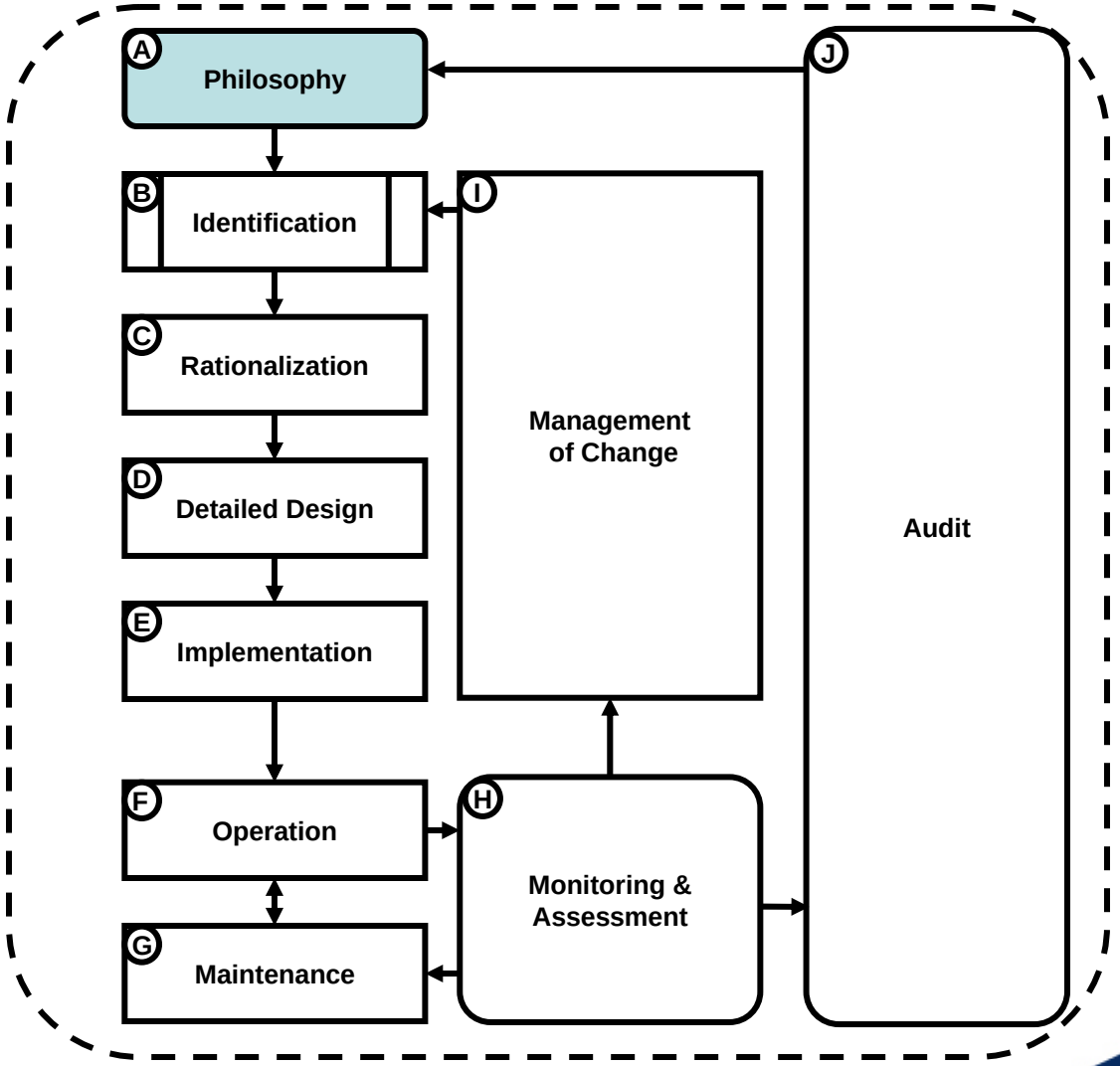
ASM – Abnormal Situation Management Consortium

EEMUA – Engineering Equipment & Materials Users' Association



Alarm Management Philosophy

- Documents the specific objectives and practices for a facility.
- Includes definitions.
- Philosophy may be used to generate an alarm system requirements specification.



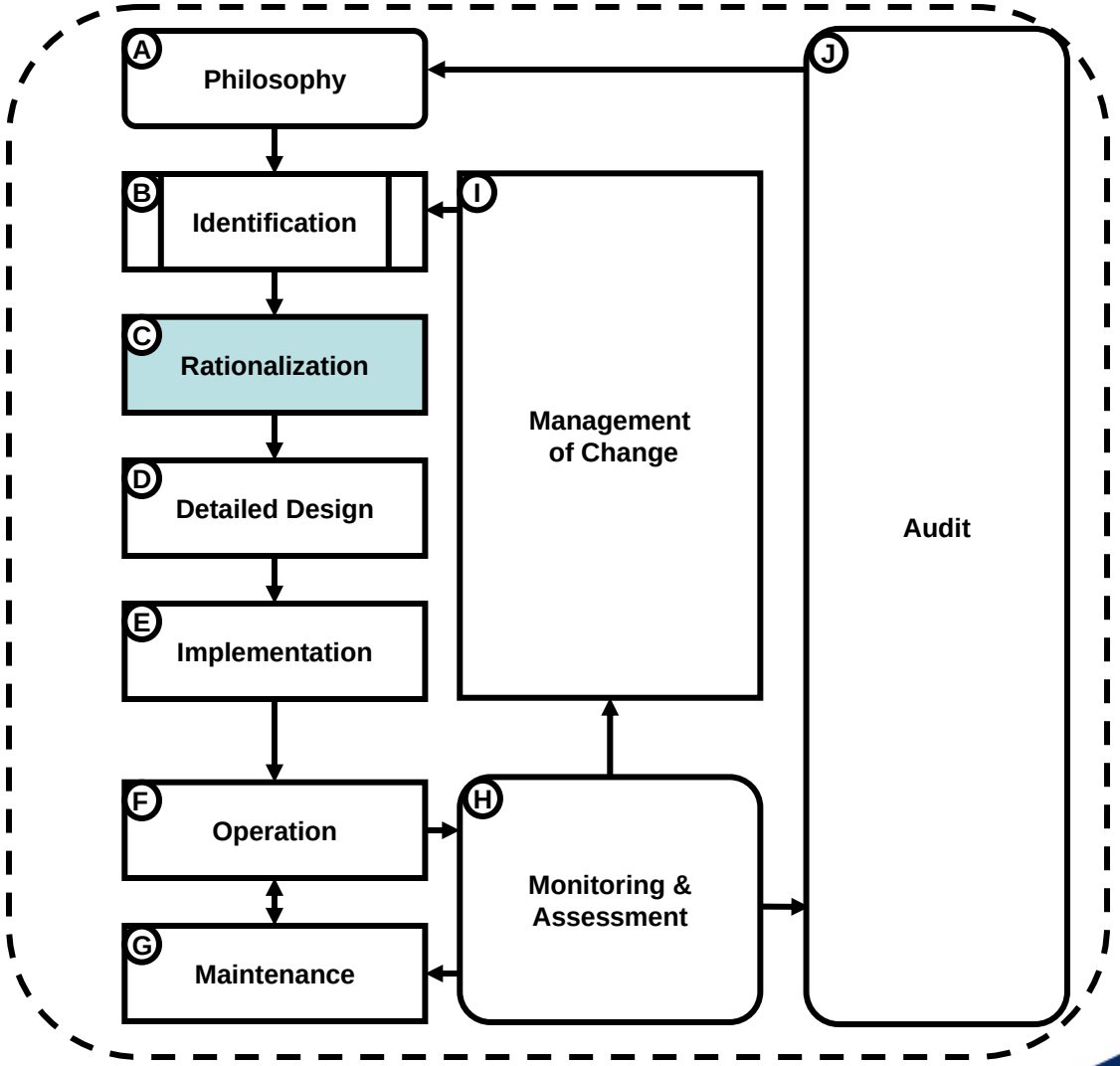
Definitions



- alarm
  - An audible or visible means of indicating to the operator an equipment or process malfunction or abnormal condition requiring an action.
- alarm management:
  - The processes and practices for determining, documenting, designing, operating, monitoring, and maintaining alarm systems.
- alarm system
  - The collection of hardware and software that detects an alarm state, transmits the indication of that state to the operator, and records changes in the alarm state.

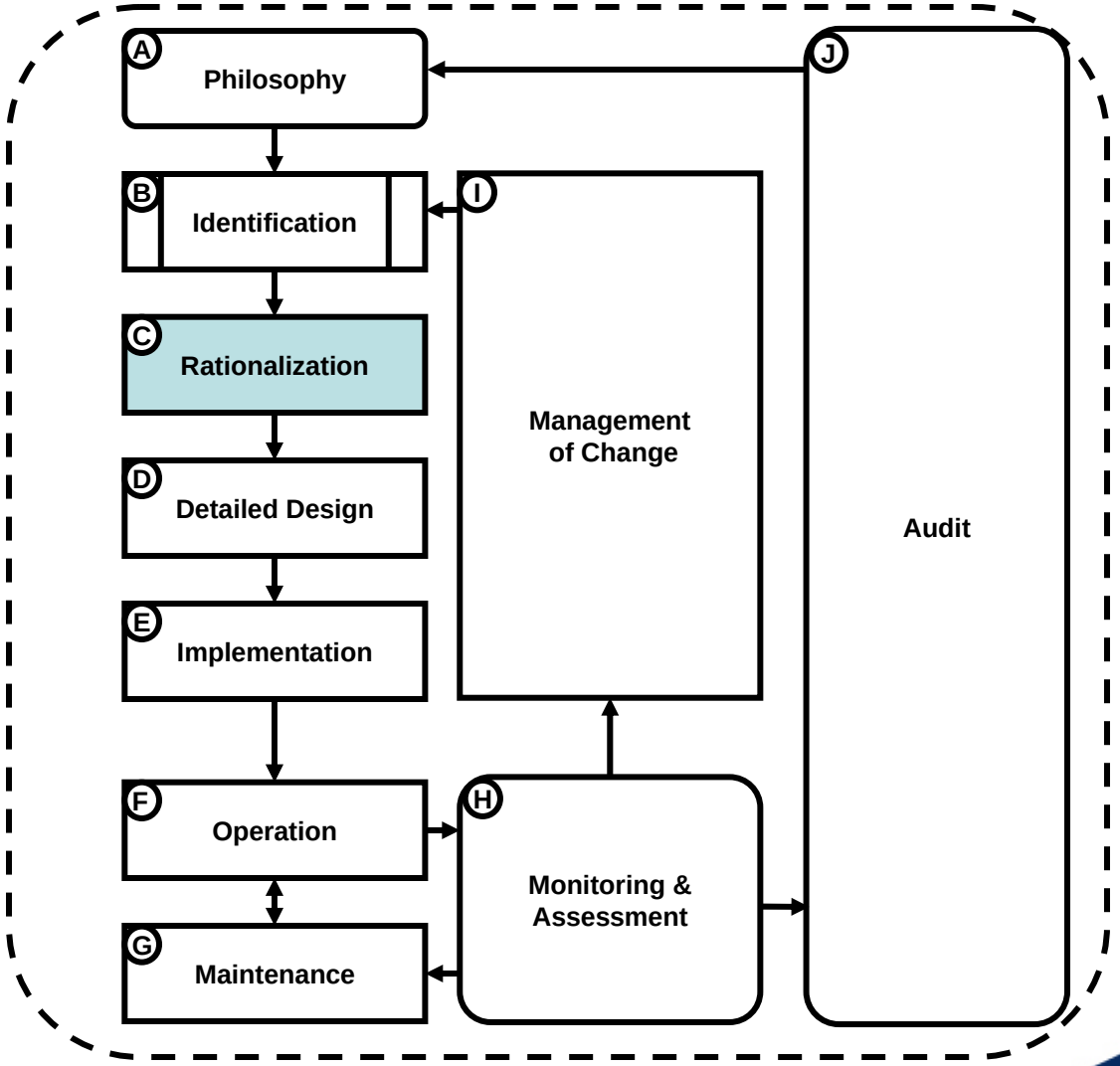
Alarm Rationalization

- Potential alarms are identified through many processes.
- Potential alarms are rationalized to documented alarms.
- Classification, prioritization, and documentation are included.



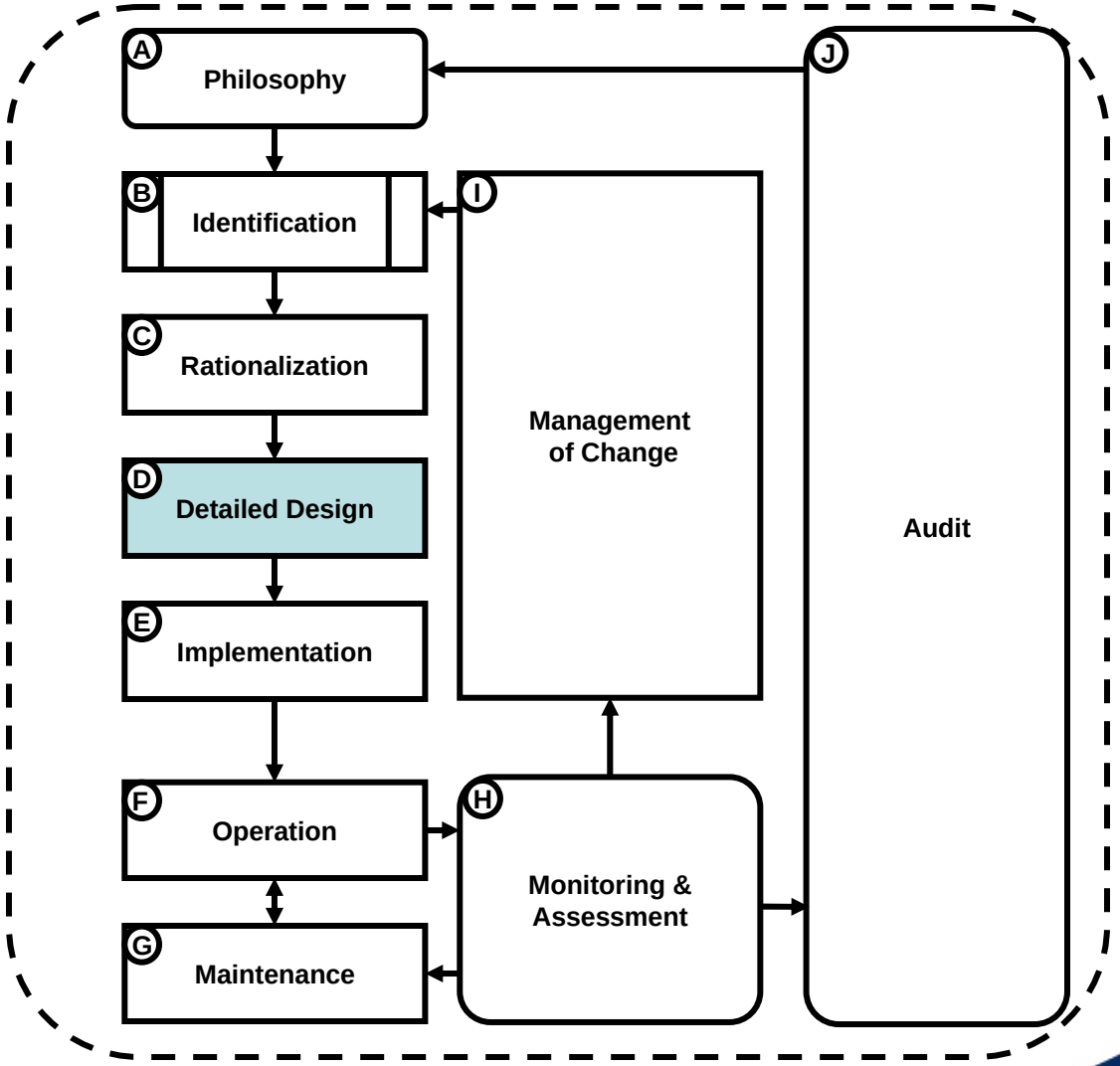
Alarm Rationalization: Problems Solved

1. Stale alarms
2. Alarms without response
3. Alarms with the wrong priority
4. Redundant alarms



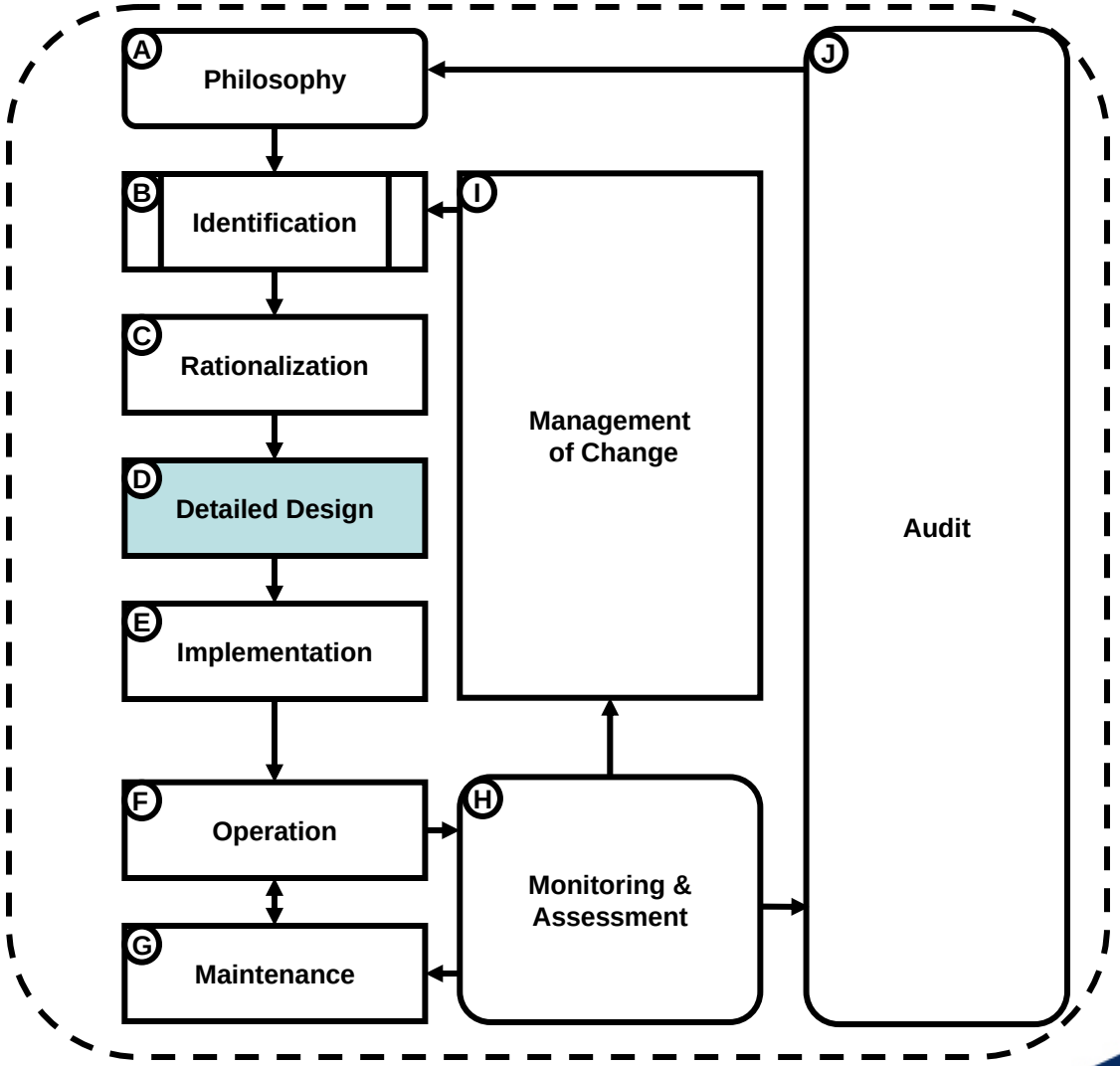
Alarm System Detailed Design

- Three parts:
  - Basic alarm design
  - HMI design
  - Advanced alarm design



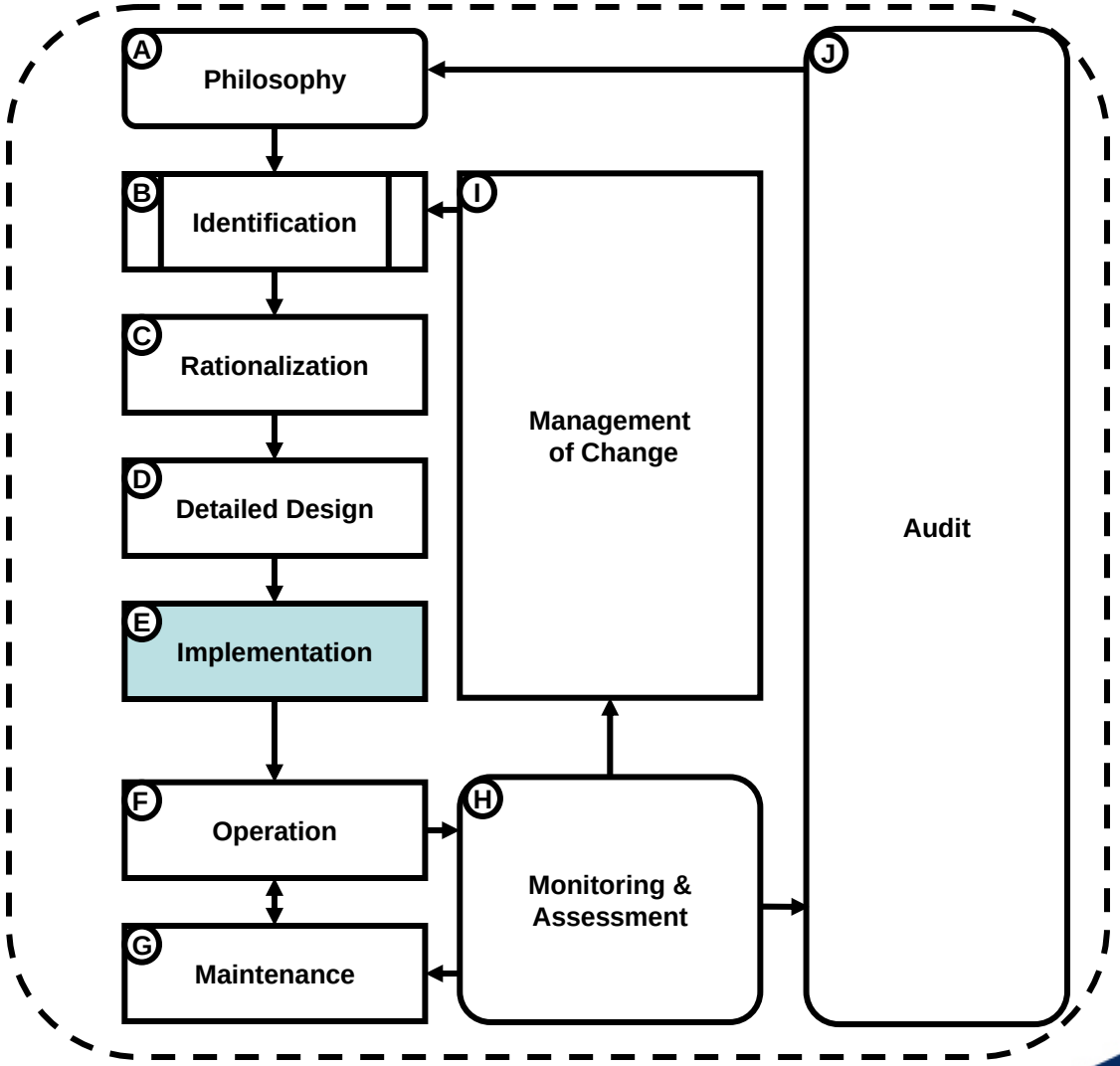
Alarm System Detailed Design: Problems Solved

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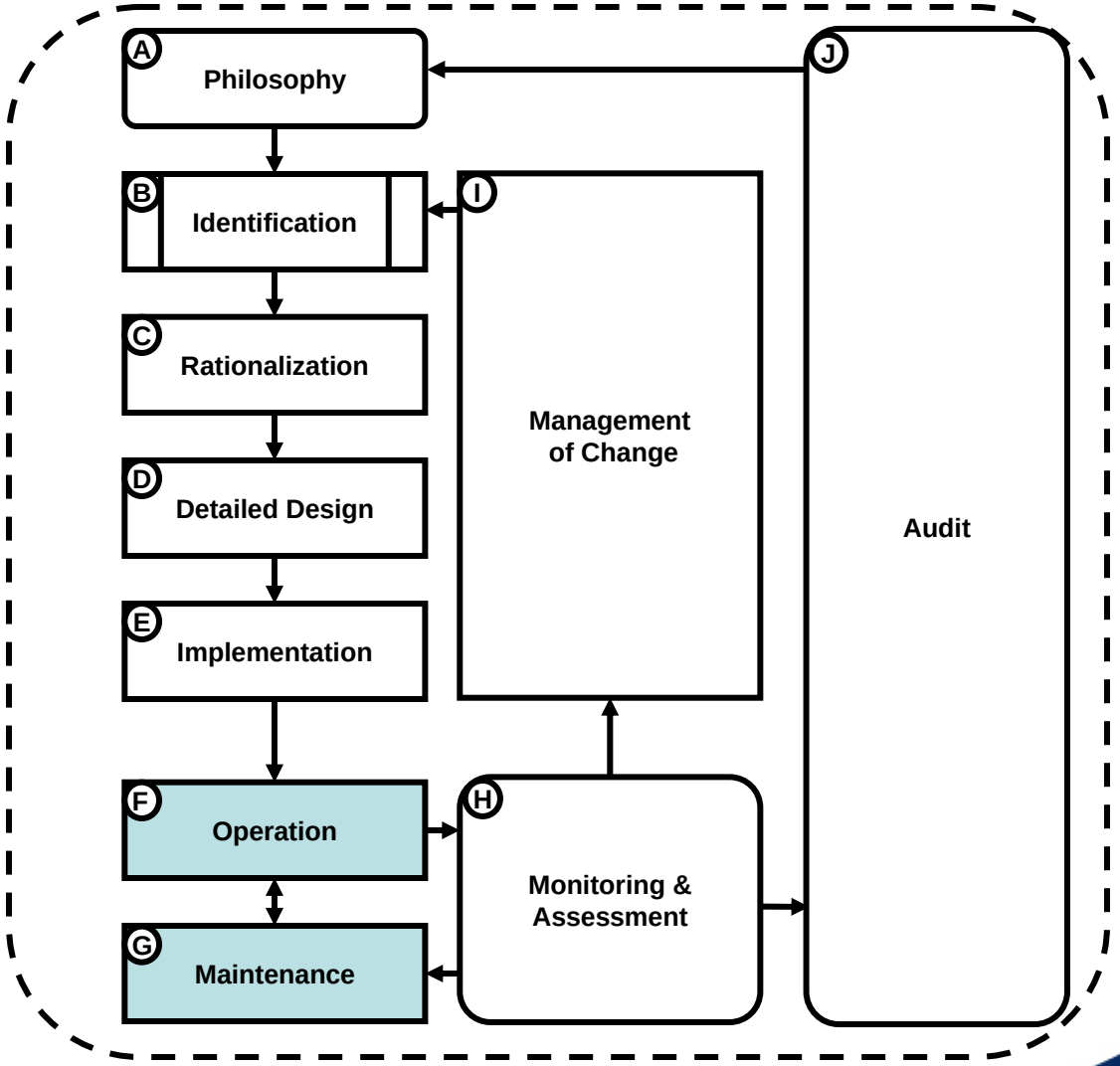
Alarm System Implementation

- Implementation is the process of putting the alarm into operation.
- Training and testing are key activities.



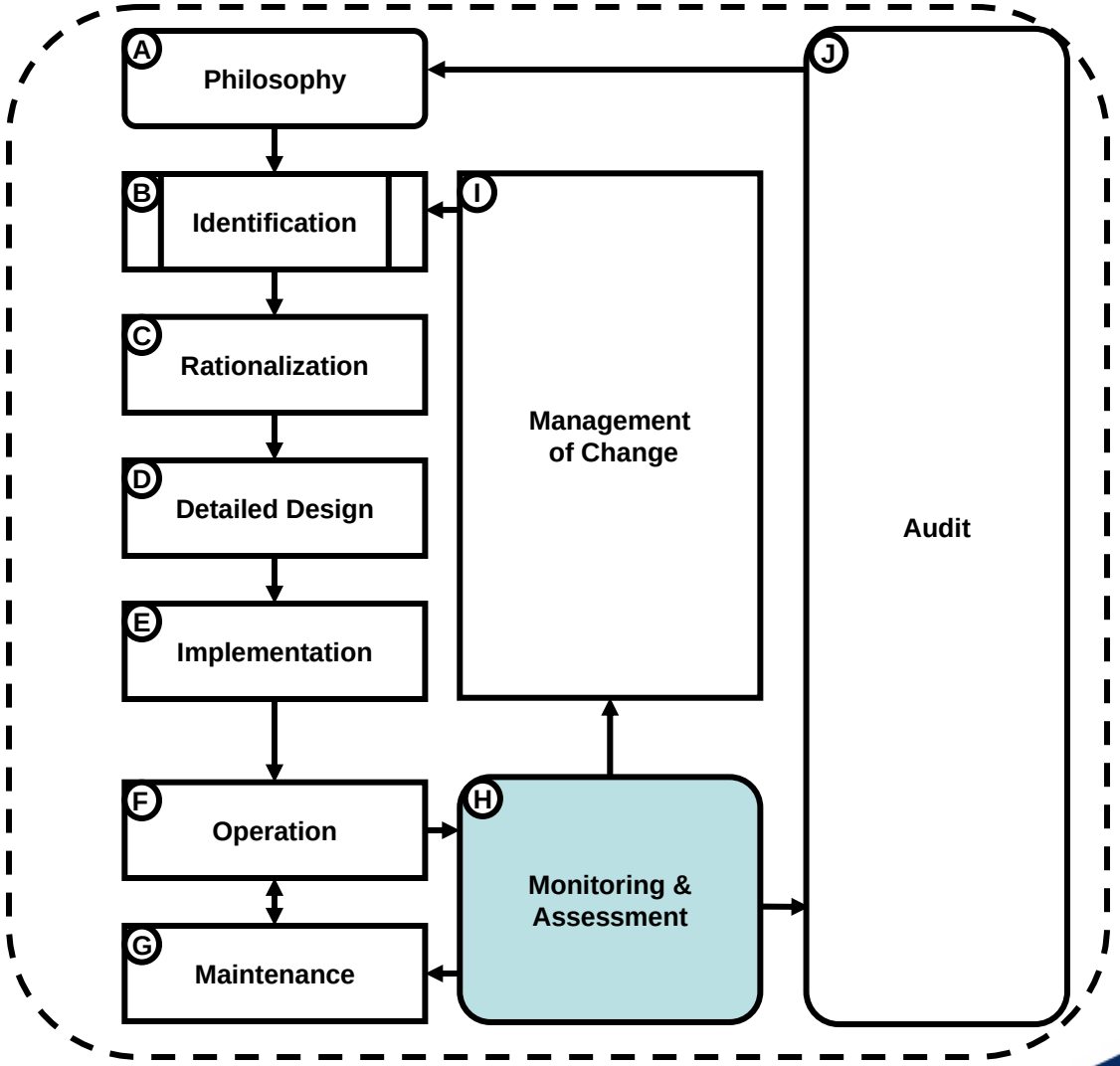
Alarm System Operation & Maintenance

- Operation is when the alarm is in service and performing its function.
- Maintenance is when the alarm is out of service for repair, replacement, or testing.



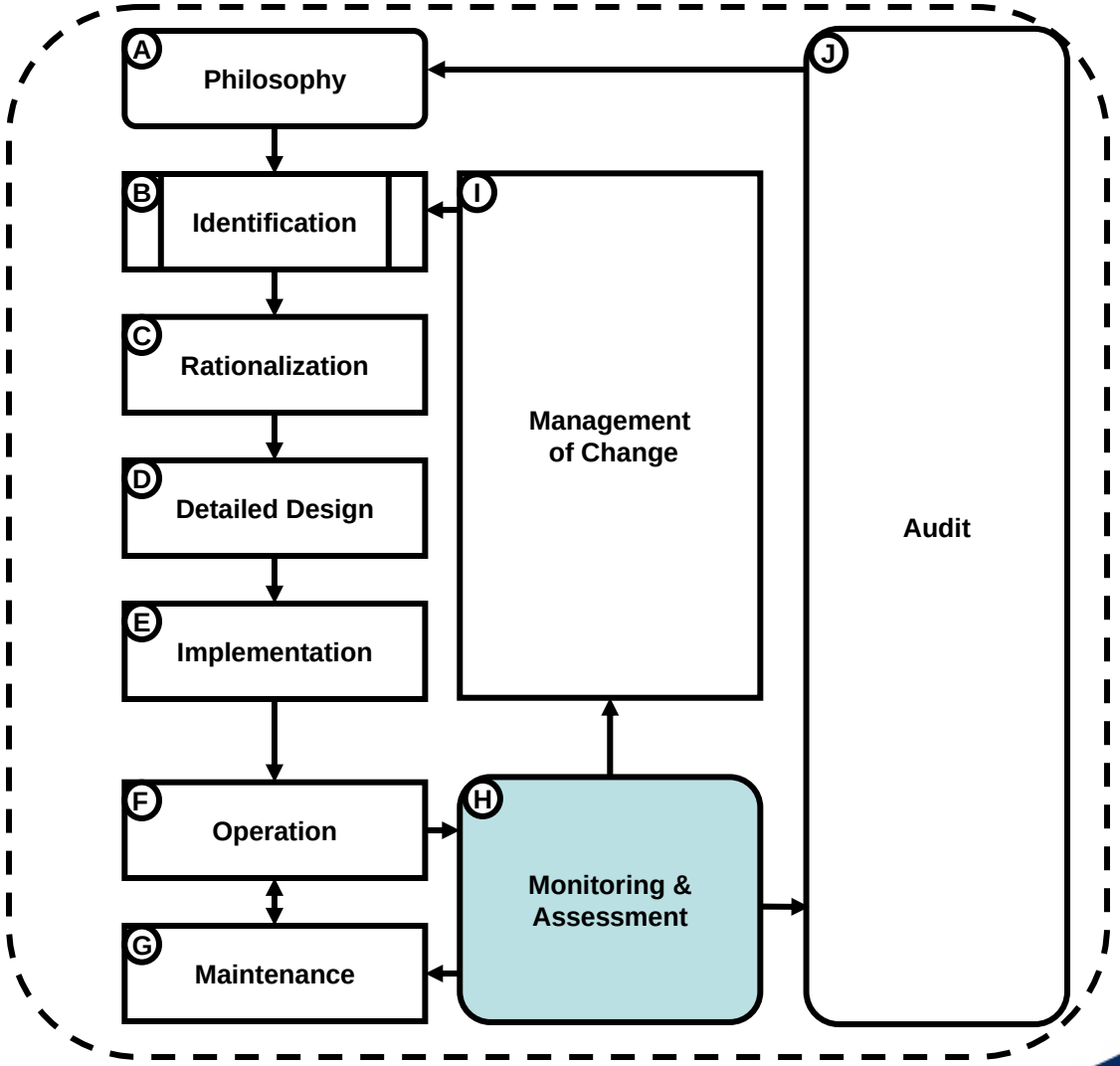
Alarm System Monitoring & Assessment

- Monitoring and Assessment is the tracking of the alarm system performance vs objectives in the Philosophy.
- An unmonitored alarm system is almost always broken.

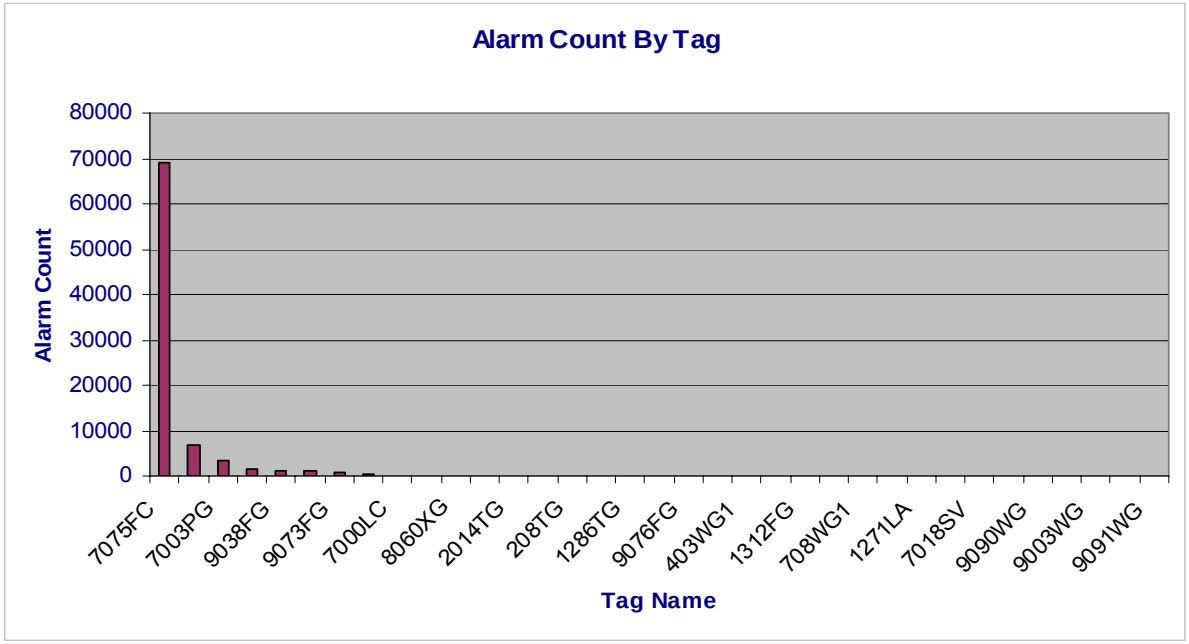


Alarm System Monitoring & Assessment: Problems Identified

1. Nuisance alarms
2. Stale alarms
3. Alarm Floods
4. Alarms with the wrong priority
5. Out-of-Service alarms
6. Redundant alarms



Example - Monitoring



Tag	Alarm Count
7075FC	68994
1019XG	6866
7003PG	3300
806FT	1440
9038FG	1304
9018TC	1112
9073FG	904
900FT	500
7000LC	140
1153FA	132
8060XG	109
9070XG	93
2014TG	90
1312XG	72
208TG	54
1017TG	51
1286TG	40

- 80/20 Rule and then some: A few points cause most alarms.
- Monitoring is the key to alarm management.

SP18 Status

- ISA Standards & Practices Committee 18 Instrument Signals and Alarms
- Current project: Standard for Management of Alarm Systems
- Start date: October, 2003
- Issue date: October, 2008?
- Very active committee with broad experience.
- Working in connection with:
  - NAMUR 102
  - ASM
  - EEMUA 191
  - ISA SP 84
  - ISA SP 101

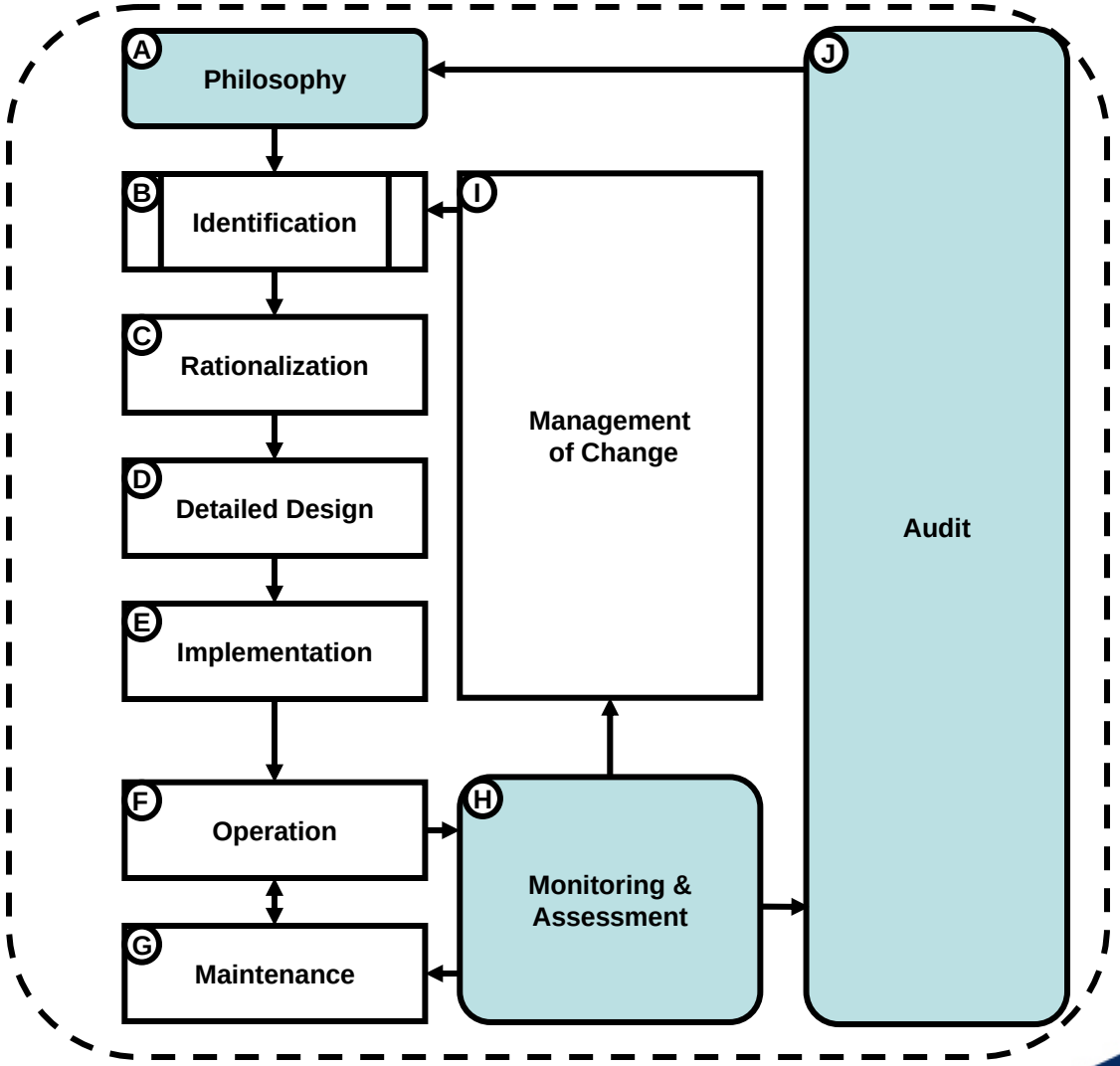
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Getting Started

- Develop a Philosophy.
- Install a monitoring package.
- Benchmark your system.
- Don’t start improvement with out a measurement.



Success:
It Can Be Done!

- Few alarms.
- Clearly prioritized and presented to the operator.
- Each with a needed action.
- Each action is taken.
- Alarms aid the operator in an upset.
- The system is monitored so performance is maintained.



References

- ISA draft standard S18.00.02 Management of Alarm Systems for the Process Industries
- Alarm Management: Seven Effective Methods for Optimal Performance
- EEMUA 191 Alarm Systems: A Guide to Design, Management and Procurement

